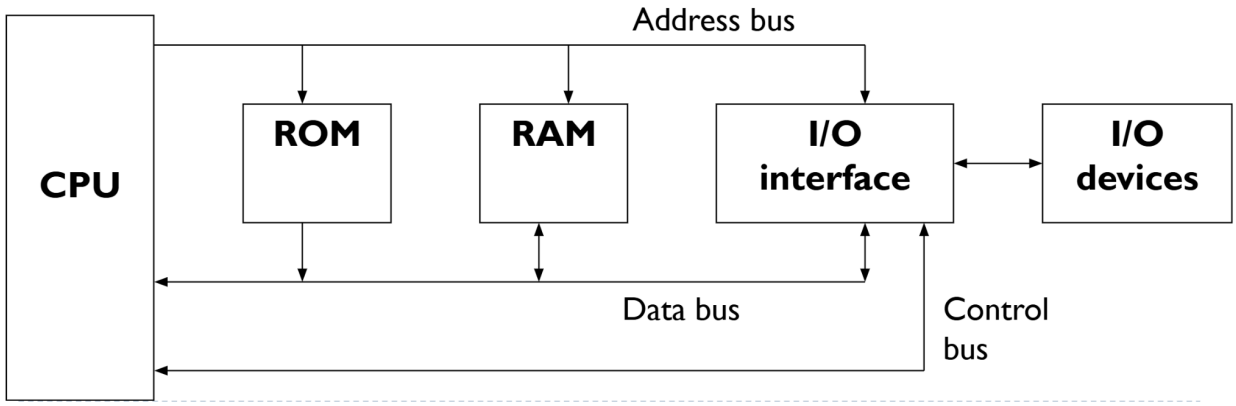
Components of Microcomputer:

- CPU (Microprocessor)
- Memory (RAM, ROM etc.)
- I/O
- **System Buses:**

- Address bus
- Data bus
- Control bus

Control Bus	Address Bus	Data Bus
• Control Bus is Bi-directional. • It carries timing and control signals generated by the CPU that are used to synchronize operation of the individual microcomputer elements. • It can carry many different signals. For e.g. - I/O Read - I/O Write - Interrupt - Memory read - Memory write	• Carries memory address of the instructions which are to be executed • Information transfer takes place **from the processor to the memory or I/O** elements. • That is why the address bus is **Unidirectional** • **The number of locations that the CPU can address is determined by the size of address bus** **For example : Microprocessor with 32 bit address bus can address 2^32 memory locations**	• It is used to carry data. • It is **bidirectional**. That means data can flow in both to or from the microprocessor. • The size of the data bus varies from one microprocessor to another. • **Usually matches the word length of the microprocessor** **For example : Microprocessor with 32 bit data bus can transmit 32 bits of data at a time**

Diagram:

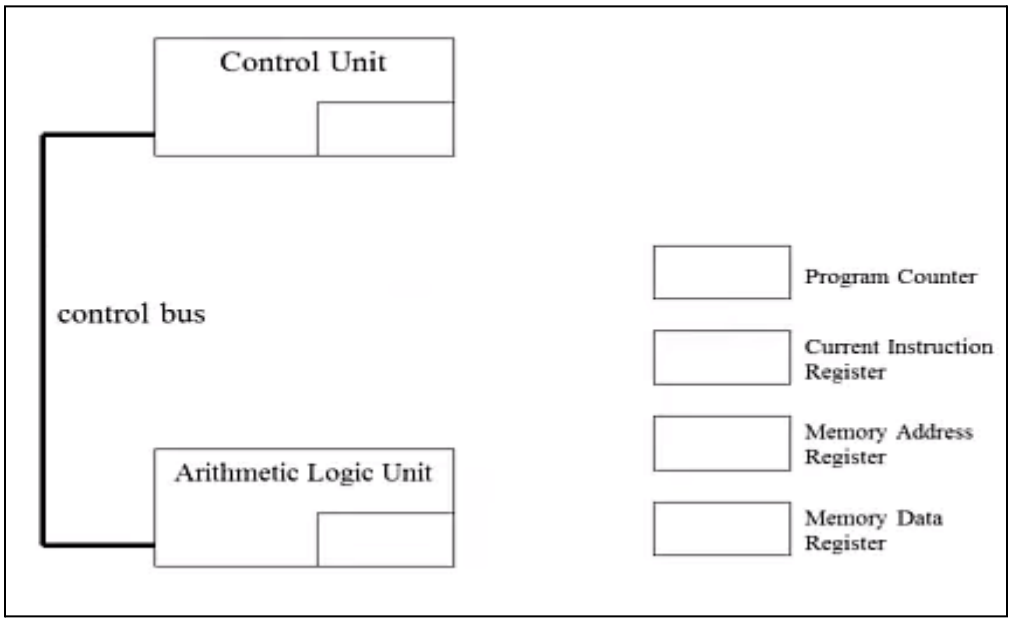


RAM	ROM
1. Can be read and written to anytime by the CPU	1. Can only be read by the CPU
2. It is a volatile memory, contents of the memory are erased when the power of the computer is turned off	2. It is a non volatile memory, contents of the memory are NOT erased when the power of the computer is turned off
3. It is not pre-loaded with any data only based on necessity we load data associated with a program in it	3. It is pre-loaded with data and software that never changes

Memory-Microprocessor-Memory:

Only 2 operations take place:

1. **READ :** When data is read from a selected memory location or an I/O port by the processor
2. **WRITE :** When data is written onto memory location or an I/O port by the processor



Fetch and Execution Cycle:

FETCH : Take in binary-coded instructions/ data from memory

DECODE : Analyze or make sense of the instructions

EXECUTE : Carry out the instructions, controls overall operation of the computer

Fetch
For Data Only
Suppose, fetching only values like - **5,6, 1001, A, B, 90AB, 888**
Necessary Registers [4]: **PC [Program counter] :** Holds the address of the next instruction to be executed **IR [Instruction Register] :** Holds the instruction currently being executed or decoded **MAR [Memory Address register]** : Holds memory address from where data will be fetched **MDR [Memory Data register]** : Holds the data being transferred to the memory or from the memory by the CPU

Fetching sequence:

1. Program Counter (PC) → Memory Address Register (MAR)

- The **Program Counter (PC)** holds the address of the next instruction to be fetched.
- The address from the PC is sent to the **Memory Address Register (MAR)** to access the instruction from memory.

2. MAR → Memory Data Register (MDR)

- The **MAR** holds the memory address, and the control unit sends a **memory read signal** to fetch the instruction from memory.
- The instruction or data at that address is loaded into the **Memory Data Register (MDR)**.

3. MDR → Instruction Register (IR)

- The **MDR** now contains the fetched instruction.
- The instruction is transferred to the **Instruction Register (IR)** for decoding and execution.

4. PC Increment

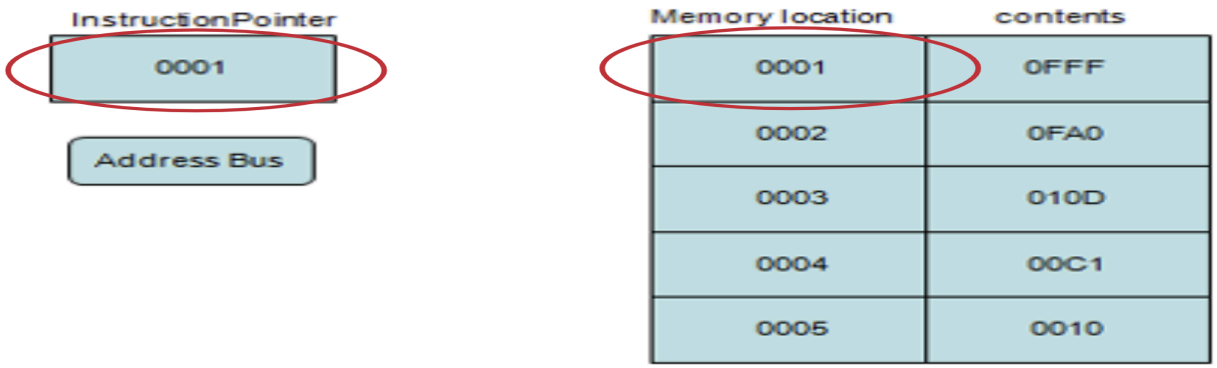
- After fetching the instruction, the **PC** is incremented to point to the next instruction in memory.

In short:

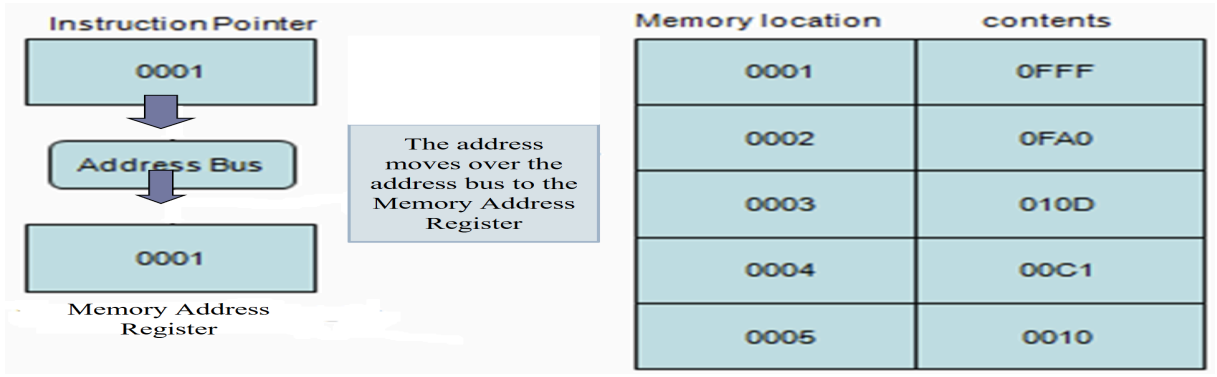
1. **PC → MAR** (Address transfer for memory fetch)
2. **MAR → MDR** (Memory fetch operation)
3. **MDR → IR** (Instruction is loaded into IR)
4. **PC is incremented** (Next instruction is prepared for fetching

Diagrammatic Representation [Data Fetch]:

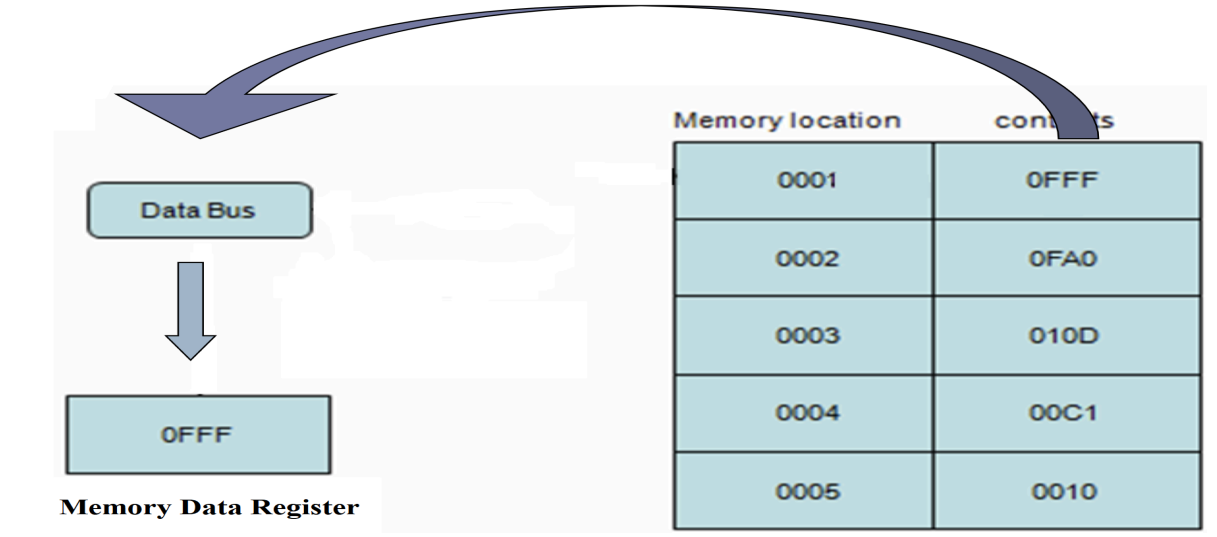
1.



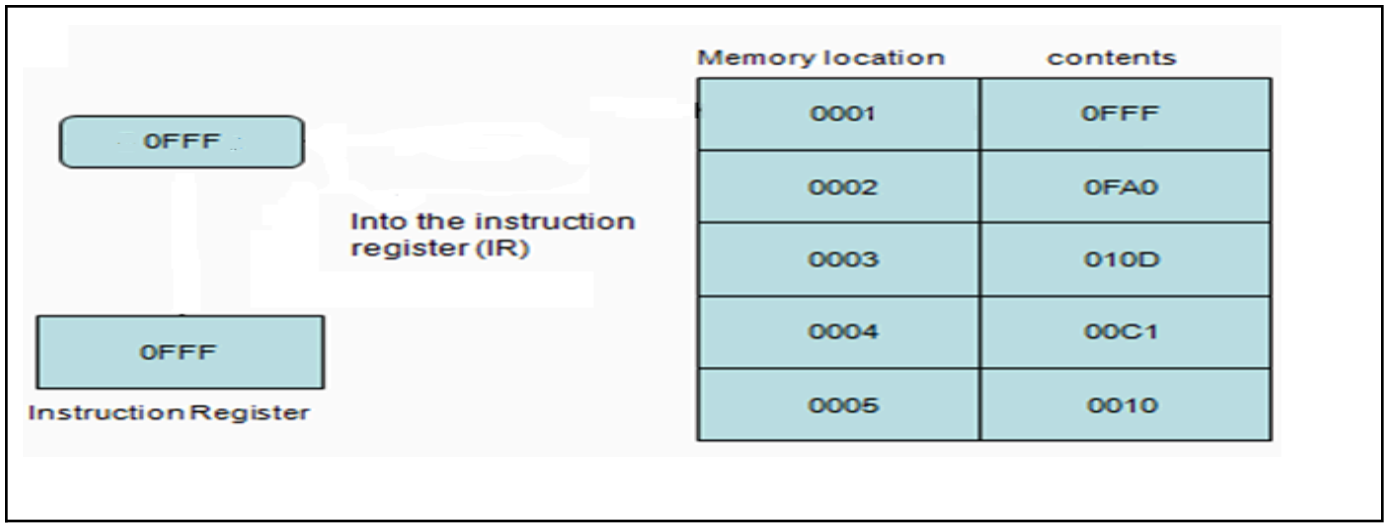
2.



3.



4.


0FFF
Into the instruction register (IR)
0FFF
Instruction Register
Memory location
contents
0001
0FFF
0002
0FA0
0003
010D
0004
00C1
0005
0010